

Fenwick Battery Storage – Site Screening Report

Selection of a 100 MW / 200 MWh BESS site

Liam McAllister · 23 January 2026 · FW-GEN-RPT-0001

1. Summary

This report documents the screening that led to the selection of the Fenwick site for a 100 MW / 200 MWh battery energy storage system. Of 14 candidate grid nodes with spare 132 kV capacity, three were shortlisted and visited in November 2025. The preferred site – a 6.2 ha pasture field immediately east of the existing **132 kV Fenwick substation** – scores best on grid proximity, planning constraints and land availability. The landowner has agreed in principle to an option.

Criterion	Weight	Fenwick	Site B (Oakhurst)	Site C (Brayton Moor)
Distance to 132 kV substation	25 %	5 (adjacent, ~250 m)	3 (2.1 km)	4 (900 m)
Indicative grid headroom	20 %	4	4	2
Planning constraints (flood, landscape, heritage)	20 %	4	3	4
Distance to nearest dwelling	10 %	4 (310 m)	2 (140 m)	5 (650 m)
Access for HGV / abnormal loads	10 %	4	4	2
Land availability	15 %	5 (willing landowner)	2	3
Weighted score		4.40	3.05	3.35

Table 1 – Multi-criteria scoring (1 = poor, 5 = very good)



The existing 132 kV Fenwick substation seen from the proposed site entrance

2. Site description

The site is a level, improved grassland field (grade 3b agricultural land) at around 45 m above sea level. It is bounded by mature hedgerows on three sides and by the substation compound to the west. Access is from the C-class road via an existing farm track, which would need widening to 6 m over 180 m.

3. Constraints

- Flood zone 1 (low probability) – a small surface water flow path along the northern hedge.
- No statutory landscape or ecological designations within 2 km; a local wildlife site 1.4 km south.
- One listed farmhouse 750 m north-east; visual impact limited by hedgerows and topography.
- Public footpath along the eastern boundary – to be retained with a 10 m buffer.
- Overhead 132 kV line crosses the south-western corner – exclusion corridor of 2 × 15 m.

4. Grid

Public network capacity data and an informal discussion with NordGrid Transmission in October 2025 indicate that a 100 MW import/export connection at Fenwick 132 kV is feasible, possibly with a bay extension. Formal confirmation can only come with the grid connection offer.

5. Recommendation

Proceed with Fenwick: negotiate a land option and prepare the grid connection application. Development budget request to the Investment Committee (see Project Pipeline & Bids). Project lead: Kwame Mensah.